

20.1 INTRODUCTION

Competition make businessmen compel to select best strategy to survive in the business world. It has been proven that the strategy taken by other organization or individual can dramatically affect the outcome of our decision. In the T.V. industry for example, the strategies of introducing new model with certain improvement in the facility can make dramatic effect on the profit of the other competitors (makers). These days, a successful businessman always makes major decision considering the possible courses of action available to an opponent involving in the same business.

'Game Theory' is a way to consider the impact of the courses of action of others on our strategies and outcomes. In other words, it is a peculiar type of decision theory in which one's choice of strategy is determined after considering the possible alternatives available to an opponent playing the same game, rather than just by the possibilities of several outcomes.

The word 'game' is not just a kind of sport but considered in broad sense as a competitive situation, a kind of conflict in which some body must win and some body must lose. Game theory assists how optimal strategies are formulated in conflict situation.

20.2 BASIC TERMINOLOGIES

Players: The autonomous decision makers in the game are called players. They may be individual groups of individual or organization. In some of the particular cases the nature might be treated as the player.

Two-person game: A game is said to be two-person game if only two players are involved in it. If more players are involved in a game the game is termed as n-person game.

Strategies: Strategies are simply decision alternatives taken by a player, while playing a game a player always selects the best strategy to optimize his gains or losses.

Optimal strategy: Optimal strategy means the complete plan adopted by a player, which optimizes his gains or losses without knowing the competitors plans.

Value of the game: Value of the game is the expected outcome per play in the game.

Pure strategy: Pure strategy refers to the campaign plan for the entire game taking into consideration of possible response to every eventuality.

Mixed strategy: Mixed strategies are those employed strategies in the game in which the interest of the players are not conceivable. In such a case players have to select two or more strategies at a time according to predetermine probabilities.

Saddle point: It is the point where maximin value equals to minimax value. The saddle point is the value of game. Once saddle point exists for a game, the optimal strategy for two players can be identified.

Two person zero sum game: It is the game involving only two players, in which the gain of one player equals to loss of other.

$$p_1 = \frac{0 - (-1)}{(3+0) - (-1-1)} = \frac{1}{5} \text{ and } p_2 = 1 - \frac{1}{5} = \frac{4}{5}$$

$$q_1 = \frac{0 - (-1)}{(3+0) - (-1-1)} = \frac{1}{5} \text{ and } q_2 = 1 - \frac{1}{5} = \frac{4}{5}$$

$$\text{and } V = \frac{3 \times 0 - (-1) \times (-1)}{3+0 - (-1-1)} = -\frac{1}{5}$$

Hence optimal strategy of player A = $\left(\frac{1}{5}, \frac{4}{5}\right)$ and optimal strategy of player B = $\left(\frac{1}{5}, \frac{4}{5}\right)$ and

$$\text{value of the game } V = -\frac{1}{5}.$$

Calculus Method

This method is based on the partial derivative of expected payoff function, which is described below.

We derive expected pay-off function $E(p, q)$ as follows by applying this method for the problem, discussed above:

$$E(p, q) = \{(\text{Expected pay off to player A when player B uses strategy } b_1) \times \text{Probabilities of player B using strategy } b_1\} + \{(\text{Expected pay-off to player A when player B uses strategy } b_2) \times \text{Probability of player B using strategy } b_2\}$$

Using this method for the above problem, we get

$$E(p, q) = [3p_1 + (-1)(1 - p_1)] \times q_1 + [(-1)p_1 + 0(1 - p_1)] \times (1 - q_1)$$

$$\text{or } E(p, q) = (3p_1q_1 - q_1 + p_1q_1) + (-p_1 + p_1q_1)$$

$$\text{or } E(p, q) = 4p_1q_1 - q_1 - p_1 + p_1q_1$$

$$\therefore E(p, q) = 5p_1q_1 - q_1 - p_1$$

Differentiating partially $E(p, q)$ with respect to p_1 and q_1 and equate to zero, we get

$$\frac{\partial E}{\partial p_1} = 5q_1 - 1$$

$$0 = 5q_1 - 1$$

$$\therefore q_1 = \frac{1}{5} \text{ and } q_2 = 1 - \frac{1}{5} = \frac{4}{5}$$

Again

$$\frac{\partial E}{\partial q_1} = 5p_1 - 1$$

$$0 = 5p_1 - 1$$

$$\therefore p_1 = \frac{1}{5} \text{ and } p_2 = 1 - \frac{1}{5} = \frac{4}{5}$$

Substituting the values of p_1 and q_1 in expected pay-off function, we get

$$\text{Value of the game, } V = 5 \times \frac{1}{5} \times \frac{1}{5} - \frac{1}{5} - \frac{1}{5} = -\frac{1}{5}$$

Hence optimal strategy of player A = $\left(\frac{1}{5}, \frac{4}{5}\right)$ and optimal strategy of player B = $\left(\frac{1}{5}, \frac{4}{5}\right)$ and value of the game

$$V = -\frac{1}{5}.$$

Matrix Method

This method is based on the principle of matrix multiplication, adjoint matrix and cofactor matrix. In this method, optimal strategies of both players A and B and the value of the game are obtained by using following formula.

$$\text{Player A's optimal strategy} = \frac{(1 \ 1)P_{\text{adj}}}{(1 \ 1)P_{\text{adj}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

$$\text{Player B's optimal strategy} = \frac{(1 \ 1)P_{\text{cof}}}{(1 \ 1)P_{\text{adj}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}$$

and, Value of the game = (Player A's optimal strategies) $\times$ (Pay-off matrix) $\times$ (Player B's optimal strategies)
 where, P_{adj} = adjoint matrix
 P_{cof} = cofactor matrix

By applying this method for the problem, discussed above, we get

$$\text{Pay-off matrix} = \begin{pmatrix} 3 & -1 \\ -1 & 0 \end{pmatrix}$$

$$P_{\text{adj}} = \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$$

$$P_{\text{cof}} = \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\therefore \text{Player A's optimal strategies} = \frac{(1 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}}{(1 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}} = \frac{(1, 4)}{5} = \left(\frac{1}{5} \ \frac{4}{5} \right)$$

Thus, we conclude that player A will employ strategy A_1 with probability $\frac{1}{5}$ and A_2 with probability $\frac{4}{5}$.

$$\text{In the similar way, player B's optimal strategies} = \frac{(1 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}}{(1 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}} = \frac{(1, 4)}{5} = \left(\frac{1}{5} \ \frac{4}{5} \right)$$

Thus, we conclude that player B will employ strategy B_1 with probability $\frac{1}{5}$ and B_2 with probability $\frac{4}{5}$.

$$\begin{aligned} \text{And value of the game, } V &= \left(\frac{1}{5} \ \frac{4}{5} \right) \begin{pmatrix} 3 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1/5 \\ 4/5 \end{pmatrix} \\ &= \left(\frac{3}{5} - \frac{4}{5} \right) \begin{pmatrix} 1/5 \\ 4/5 \end{pmatrix} \\ &= \left(-\frac{1}{5} \right) \begin{pmatrix} 1/5 \\ 4/5 \end{pmatrix} = \left(-\frac{1}{25} - \frac{4}{25} \right) = \left(-\frac{5}{25} \right) = -\frac{1}{5}. \end{aligned}$$

The matrix method is useful for the square matrix of higher order.

Graphical Method

This method is developed to reduce than higher order of matrix into 2×2 at once. However, this is used only when at least one of the players in two-person zero sum game has only two pure strategies. In other words, the game matrix should be order of either $2 \times n$ or $m \times 2$.

Let us consider a game of $2 \times n$ pay-off matrix without saddle point as follows.

Player A's strategy	Player B's strategy			
	B_1	B_2	...	B_n
A_1	a_{11}	a_{12}	...	a_{1n}
A_2	a_{21}	a_{22}	...	a_{2n}

Here, player A has two strategies A_1 and A_2 with their probability of selection p_1 and p_2 such that $p_1 + p_2 = 1$ and $0 \leq p_1 \leq 1$.

Expected pay-off of player A will be calculated given that B employs the strategies $B_1, B_2, \dots, B_n$, which are tabulated as follows.

B's pure strategies	Expected payoff of player A
B_1	$a_{11} p_1 + a_{21} p_2$
B_2	$a_{12} p_1 + a_{22} p_2$
$\vdots$	$\vdots$
B_n	$a_{1n} p_1 + a_{2n} p_2$

Now, the straight lines representing the A's expected pay-off values are plotted in the graph. According to maximin strategy of player A, the highest point on the lower boundary of these lines in a graph will provide maximum expected pay off among the minimum expected payoffs and the values of p_1 and p_2 .

It should be noted that graphical method is used to reduce the given game matrix to a 2×2 game. The considered problem discussed above is of 2×2 game. So, instead of graphical method, we use any of the earlier discussed method to solve it, which we have already illustrated. However, one more method linear programming method will be discussed later.

Here, let's illustrate the graphical method considering the following game of 2×3 . The payoff is given from player A's perspective.

Player A's strategy	Player B's strategy		
	B_1	B_2	B_n
A_1	1	3	11
A_2	8	5	2

Solution

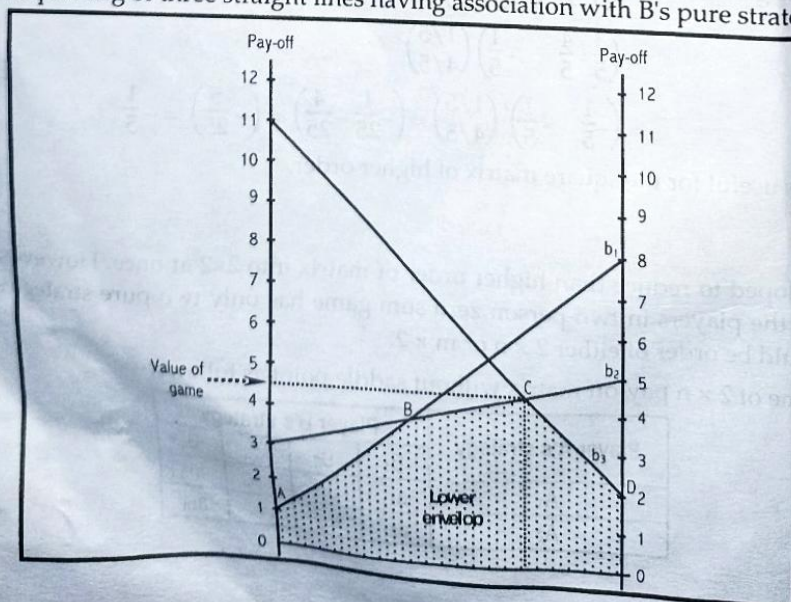
The given game does not have a saddle point, as maximin value is not equal to minimax value. Let p_1 and p_2 be the probabilities of selecting strategies A_1 and A_2 by player A such as $p_1 + p_2 = 1$, $0 \leq p_1 \leq 1$. Then, expected pay-offs of player A will be as follows.

B's pure strategies	Expected payoff of player A
B_1	$p_1 + 8p_2$
B_2	$3p_1 + 5p_2$
B_3	$11p_1 + 2p_2$

Figure 20.2 shows the plotting of three straight lines having association with B's pure strategies.

FIGURE 20.2

Graph of expected pay-off and pure strategies (to show lower envelop)



The lower envelope is ABCD as shown in the Figure 20.1. Player A will choose the possible strategies in such a way that he will have maximum expected pay-off. He/she finds maximum expected payoff at point C, which is determined by the lines representing pure strategies B_2 and B_3 of player B. Now, the given game of order 2×3 has been reduced to the order of (2×2) as given below

Player A	Player B	
	B_2	B_3
A_1	3	11
A_2	5	2

The optimal strategies of player A, now can be obtained by equating two lines representing pure strategies B_2 and B_3 .

$$3p_1 + 5p_2 = 11p_1 + 2p_2$$

$$\therefore p_1 = \frac{3}{11} \text{ and } p_2 = 1 - p_1 = \frac{8}{11}$$

Substituting the value of p_1 and p_2 in any of the above lines, we get the value of game (expected gain to player A)

$$\therefore \text{Value of game} = 3 \times \frac{3}{11} + 5 \times \frac{8}{11} = \frac{49}{11}$$

Now, let q_1 and q_2 be the probabilities of selecting strategies B_2 and B_3 by player B such that $q_1 + q_2 = 1$, $0 \leq q_1 \leq 1$. Then, optimal strategies of player B can be obtained by equating two lines representing two strategies A_1 and A_2 of player A. i.e.

$$3q_1 + 11q_2 = 5q_1 + 2q_2$$

$$\therefore q_1 = \frac{9}{11} \text{ and } q_2 = 1 - q_1 = 1 - \frac{9}{11} = \frac{2}{11}$$

Substituting the values of q_1 and q_2 in any of the above two lines, we get the value of game (expected loss to player B)

$$\therefore \text{Value of game } V = 3 \times \frac{9}{11} + 11 \times \frac{2}{11} = \frac{49}{11}$$

$$\text{Hence, Optimal strategies of A} = \left(\frac{3}{11}, \frac{8}{11} \right).$$

$$\text{Optimal strategies of B} = \left(0, \frac{9}{11}, \frac{2}{11} \right)$$

$$\text{and Value of the game, } V = \frac{49}{11}.$$

We may use algebraic method or arithmetic method for the reduced matrix of order 2×2 . However equating two lines representing pure strategies is more common in graphical method of solving game matrix of mixed strategy.

Again, let's illustrate the graphical method for a game of $m \times 2$ payoff matrix without saddle point. Hence let's consider the following game of 5×2 . The payoff is given from player A's perspective.

Player A's strategy	Player B's strategy	
	B_1	B_2
A_1	-6	7
A_2	4	-5
A_3	-1	-2
A_4	-2	5
A_5	7	-6

Solution

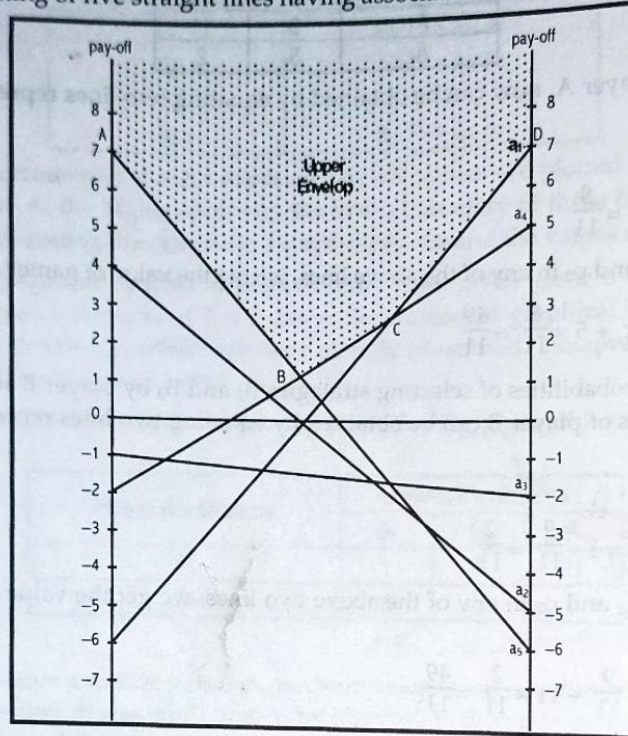
The given game does not have a saddle point. Let q_1 and q_2 be the probabilities of selecting strategies B_1 and B_2 by player B such that $q_1 + q_2 = 1$, $0 \leq q_1 \leq 1$. Then, expected payoff of player B will be as follows:

A's pure Strategies	Expected payoff of player B
A_1	$-6q_1 + 7q_2$
A_2	$4q_1 - 5q_2$
A_3	$-q_1 - 2p_2$
A_4	$-2q_1 + 5q_2$
A_5	$7q_1 - 6q_2$

Figure 20.3 shows the plotting of five straight lines having association with A's pure strategies.

FIGURE 20.3

Graph of expected pay-off and pure strategies (to show upper envelop)



The upper envelop is ABCD as show in the figure 20.2 player B will choose the possible strategies in such a way that he will have minimum expected loss. He/she finds minimum expected loss at point B which is determined by the lines representing pure strategies A_4 and A_5 of player A.

Now the given game of order 5×2 has been reduced to the order of 2×2 as given below.

Player A	Player B	
	B_1	B_2
A_4	-2	5
A_5	7	-6

By adopting the same method as discussed above we need to equate the lines

$$-2q_1 + 5q_2 = 7q_1 - 6q_2 \text{ and } -2p_1 + 7p_2 = 5p_1 - 6p_2$$

we get,

$$q_1 = \frac{11}{20} \text{ and } q_2 = \frac{9}{20} \text{ and}$$

$$\therefore \text{Value of game} = \frac{23}{20}$$

$$\text{Hence, Optimal strategies of A} = \left(0, 0, 0, \frac{13}{20}, \frac{7}{20}\right)$$

$$\text{Optimal strategies of B} = \left(\frac{11}{20}, \frac{9}{20}\right)$$

$$\text{and Value of the game, } V = \frac{23}{20}$$

Linear Programming Method

This method is based on the linear programming problem and dual linear programming problem to understand this method; one should be capable of using simplex method or revised simplex method.

We discussed the principle of this method by considering only general 2×2 game. Let's consider a simple game of 2×2 pay-off matrix. The following table is the pay-off of matrix from player A's perspective.

A's strategy	B's strategy	
	B ₁	B ₂
A ₁	a ₁₁	a ₁₂
A ₂	a ₂₁	a ₂₂

Let's suppose that p_1 and p_2 be the probabilities of selecting strategies A₁ and A₂ by player A such that $p_1 + p_2 = 1$, $0 \leq p_1 \leq 1$. Similarly, q_1 and q_2 be for strategies B₁ and B₂ of player B. Now, expected payoff functions of both players are regarded linear constraints. Let's transform the game into linear programming from player A's perspective. Now,

B's pure strategies	Expected payoff function of player A
B ₁	$a_{11}p_1 + a_{21}p_2$
B ₂	$a_{12}p_1 + a_{22}p_2$

As the player A follows criterion of maximin strategy, his objective is to maximize the minimum gain. In other words, he gains at least the value of the game V from player B. Hence, the constraints are set up as follows.

$$a_{11}p_1 + a_{21}p_2 \geq V$$

$$a_{12}p_1 + a_{22}p_2 \geq V$$

where

$$p_1 + p_2 = 1, p_1, p_2 \geq 0$$

Dividing both side of the constraints by V (assuming V as positive), we get

$$a_{11}\frac{p_1}{V} + a_{21}\frac{p_2}{V} \geq 1$$

$$a_{12}\frac{p_1}{V} + a_{22}\frac{p_2}{V} \geq 1$$

$$\text{and } \frac{p_1}{V} + \frac{p_2}{V} = 1, \frac{p_1}{V}, \frac{p_2}{V} \geq 0$$

Replacing $\frac{p_1}{V}$ by x_1 and $\frac{p_2}{V}$ by x_2 , we get

$$a_{11}x_1 + a_{21}x_2 \geq 1$$

$$a_{12}x_1 + a_{22}x_2 \geq 1$$

$$\text{and } x_1 + x_2 = \frac{1}{V}, x_1, x_2 \geq 0$$

Now, the objective of player A is to maximize V , or minimize $\left(\frac{1}{V}\right) = Z$

Hence, the required model of LPP is as follows:

$$\min Z = x_1 + x_2$$

subject to constraint

$$a_{11}x_1 + a_{21}x_2 \geq 1$$

$$a_{12}x_1 + a_{22}x_2 \geq 1$$

$$\text{and } x_1, x_2 \geq 0$$

If we consider the game of order $m \times n$ where m strategies ($i = 1, 2, \dots, m$) are of player A and n strategies that ($j = 1, 2, \dots, n$) that of player B. Let a_{ij} be the pay-off corresponding to i^{th} row and j^{th} column of the matrix.

Then, the required model of linear programming problem is

$$\text{minimize } Z = x_1 + x_2 + \dots + x_m$$

subject to

$$a_{11}x_1 + a_{21}x_2 + \dots + a_{m1}x_m \geq 1$$

$$a_{12}x_1 + a_{22}x_2 + \dots + a_{m2}x_m \geq 1$$

:

$$a_{1n}x_1 + a_{2n}x_2 + \dots + a_{mn}x_m \geq 1$$

and $x_1, x_2, \dots, x_m \geq 0$

where

$$x_j = p_j/V \geq 0; j = 1, 2, \dots, m$$

From player B's perspective

A's pure strategies	Expected payoff function of player B
A ₁	$a_{11}q_1 + a_{12}q_2$
A ₂	$a_{21}q_1 + a_{22}q_2$

As the player B follows criterion of minimax strategy, his objective is to minimize the maximum loss. In other words, he loses at most the value of the game V as he has to offer this to player A in playing a game. Hence, the constraints are set up as follows:

$$a_{11}q_1 + a_{12}q_2 \leq V$$

$$a_{21}q_1 + a_{22}q_2 \leq V$$

where, $q_1 + q_2 = 1, q_1, q_2 \geq 0$

Dividing both sides of the constraints by V (assuming V as positive)

$$\frac{q_1}{V} + \frac{q_2}{V} \leq 1$$

$$\frac{a_{21}q_1}{V} + \frac{a_{22}q_2}{V} \leq 1$$

where, $\frac{q_1}{V} + \frac{q_2}{V} = \frac{1}{V}; \frac{q_1}{V}, \frac{q_2}{V} \geq 0$

Replacing $\frac{q_1}{V}$ by w_1 and $\frac{q_2}{V}$ by w_2 , we get

$$a_{11}w_1 + a_{12}w_2 \leq 1$$

$$a_{21}w_1 + a_{22}w_2 \leq 1$$

where $w_1 + w_2 = \frac{1}{V}; w_1, w_2 \geq 0$.

As the objective of player B is to minimize V or maximize $\left(\frac{1}{V}\right) = Z$ the required model of LPP is as follows:

$$\max Z = w_1 + w_2$$

Subject to constraint

$$a_{11}w_1 + a_{12}w_2 \leq 1$$

$$a_{21}w_1 + a_{22}w_2 \leq 1$$

and $w_1, w_2 \geq 0$

If we consider the game of order $m \times n$, then required linear programming problem is as follows:

$$\max Z = w_1 + w_2 + \dots + w_n$$

Subject to

$$a_{11}w_1 + a_{12}w_2 + \dots + a_{1n}w_n \leq 1$$

$$a_{21}w_1 + a_{22}w_2 + \dots + a_{2n}w_n \leq 1$$

:

$$a_{m1}w_1 + a_{m2}w_2 + \dots + a_{mn}w_n \leq 1$$

and $w_1, w_2, w_n \geq 0$

where, $w_j = q_j/V \geq 0; j = 1, 2, \dots, n$

Clearly, we can say this is the dual of the LPP transformed earlier. After solving the linear programming problem we need to transform to the original values to obtain the mixed strategy.

20.8 DOMINANCE RULE

In case saddle point does not exist in solving theory of games i.e. Maximin value is not equal to minimax value we need to reduce the size of the given matrix into 2×2 matrix so that we could use any method out of Algebraic, Arithmetic, Calculus and Matrix Method to arrive at the final solution.

For the reduction of given size of the matrix we have two methods VIZ: Dominance rule and Graphical Method. Graphical method is specially used for the matrix of either $m \times 2$ matrix or $2 \times n$ matrix. Matrix of larger size could be initially reduced by Dominance rule thereby we could use Graphical method when we arrive at either $m \times 2$ or $2 \times n$ sized matrix (but it is not necessary to use; we may completely use dominance rule completely until we arrive at 2×2 matrix). Graphical method was already discussed in this chapter. Here, we will describe how to use Dominance rule.

Dominance rule is based on the principle that we determine the strategy, which is preferable to some other strategy/strategies. In other words, we delete the strategy from the list, which is found to be less attractive to at least one of the existing strategies in terms of pay-off. This determination is done from the perspective of both players A and B one by one or at a same time.

From winner's perspective a certain row(strategy) is considered as a less attractive one comparing to other rows(strategies) if he/she finds each element of this row is either less than or equal to corresponding elements of other row(s). Once found as less attractive or inferior, the row is deleted from the given matrix.

From loser's perspective a certain column (strategy) is considered as a less attractive one comparing to other columns (strategies) if he/she finds each element of this column is either greater than or equal to corresponding elements of other column(s). Once found as less attractive or inferior, the column is deleted from the given matrix.

However, sometime we might not find the complete inferiority of the strategy comparing to other strategy (ies) as discussed above. In such a case we check the inferiority of the strategy to an average of two or more than other pure strategies. In case we find strict domination we delete the strategy, if not we will find the strategy itself dominates the convex linear combination of some other pure strategies, one of the pure strategies involved in the combination may be deleted. This technique also requires the perspectives of both players.

Note: In case of inability of solving the game theory even from average rule of dominance, we can use linear programming method to solve the problem.

The following two examples will illustrate the matter discussed in dominance rule.

EXAMPLE -3 Find the optimal strategies by using the dominance rule for Player A and B in the following game. Also obtain the value of the game.

A's Strategy	B's strategy		
	B1	B2	B3
A1	24	22	-8
A2	12	-6	14
A3	18	20	-8

Solution:

A's Strategy	B's strategy			Row minima
	B1	B2	B3	
A1	24	22	-8	-8
A2	12	-6	14	-6
A3	18	20	-8	-8
Column maxima	24	22	14	

Maximin

Minimax

Since maximin value is not equal to minimax value the saddle point does not exist. Hence, we are supposed to use dominance rule to reduce the size of the given matrix.

While using Dominance rule we use the perspectives of both players to identify the less attractive strategy (ies) comparing to other existed strategy (ies).

At first we use perspective of player A (we may take B as well); we find strategy A3 is less attractive or inferior to A1 as each element of A3 is less than or equal to corresponding element of A1. Hence, A3 is eliminated. Then the reduced matrix will be as follows:

A's strategy	B's strategy		
	B1	B2	B3
A1	24	22	-8
A2	12	-6	14

Again from Player B's perspective: we find Strategy B1 is less attractive or inferior to strategy B2 as each element of B1 is greater than or equal to corresponding element of B2. Hence, B1 is eliminated. Then the reduced matrix will be as follows:

A's strategy	B's strategy	
	B2	B3
A1	22	-8
A2	-6	14

To get final solution now we can use Arithmetic method:

Here, $a_{11}=22$, $a_{12}=-8$, $a_{21}=-6$, and $a_{22}=14$.

Considering P_1 and P_2 be the probabilities of employing strategies A1 and A2 by player A and q_1 and q_2 be probabilities of employing strategies B2 and B3 of player B, then

$$P_1 = \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{14 - (-6)}{(22 + 14) - (-8 - 6)} = \frac{20}{50} = \frac{2}{5}$$

$$P_2 = 1 - P_1 = 1 - \frac{2}{5} = \frac{3}{5}$$

$$\text{And } q_1 = \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{14 - (-8)}{(22 + 14) - (-8 - 6)} = \frac{22}{50} = \frac{11}{25}$$

$$q_2 = 1 - q_1 = 1 - \frac{11}{25} = \frac{14}{25}$$

and Value of the game,

$$V = \frac{a_{11}a_{22} - a_{12}a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{22 \times 14 - (-8)(-6)}{(22 + 14) - (-8 - 6)} = \frac{260}{50} = 5.2$$

Hence, the probability of selecting strategies optimally for players A and B are $(\frac{2}{5}, \frac{3}{5}, 0)$ and $(0, \frac{11}{25}, \frac{14}{25})$ respectively and Value of game is $V=5.2$.

Alternative Method

A's strategy	B's strategy	
	B2	B3
A1	22	-8
A2	-6	14

$$14 + 6 = 20$$

$$22 + 8 = 30$$

$$P(A_1) = \frac{20}{20 + 30} = \frac{20}{50} = \frac{2}{5}$$

$$P(A_2) = \frac{30}{20 + 30} = \frac{30}{50} = \frac{3}{5}$$

$$14 + 8 = 22$$

$$P(B_2) = \frac{22}{22 + 28} = \frac{22}{50} = \frac{11}{25}$$

$$22 + 6 = 28$$

$$P(B_3) = \frac{28}{22 + 28} = \frac{28}{50} = \frac{14}{25}$$

$$\text{A's optimal strategy} = \left(\frac{2}{5}, \frac{3}{5}, 0\right)$$

$$\text{A's optimal strategy} = \left(0, \frac{11}{25}, \frac{14}{25}\right)$$

$$\text{Value of game} = 22 \times \frac{2}{5} + (-6) \times \frac{3}{5} = \frac{26}{5} = 5.2$$

EXAMPLE -4

Find the optimal strategies by using the dominance rule for Player A and B in the following game. Also obtain the value of the game.

Solution:

A's Strategy	B's strategy		
	B1	B2	B3
A1	2	3	9
A2	4	9	7
A3	3	7	12

A's Strategy	B's strategy			Row minima
	B1	B2	B3	
A1	2	3	9	2
A2	4	9	7	4
A3	3	7	12	7
Column maxima	4	9	12	

Minimax
Maximin

Since maximin value is not equal to minimax value the saddle point does not exist. Hence, we are supposed to use dominance rule to reduce the size of the given matrix.

From Player B's perspective: we find Strategy B3 is less attractive or inferior to strategy B1 as each element of B1 is greater than (or equal to) corresponding element of B3. Hence, B3 is eliminated. Then the reduced matrix will be as follows:

Player A's strategy	Player B's strategy	
	B1	B2
A1	2	3
A2	4	9
A3	3	7

Now, we use perspective of player A: we find strategy A1 is less attractive or inferior to both A2 and A3 as each element of A1 is less than or equal to corresponding elements of A2 and A3. Hence, A1 is eliminated. Then the reduced matrix will be as follows:

Player A's strategy	Player B's Strategy	
	B1	B2
A2	4	9
A3	3	7

Let us check whether a saddle point exists or not

Player A's strategy	Player B's strategy		Row minima
	B1	B2	
A2	4	9	4
A3	3	7	3
Column Maxima	4	9	

Minimax
Maximin

Since maximin value = minimax value, the saddle point exists.
Hence, optimal strategy is (A2, B1)
And Value of the game (V) = 4.

EXAMPLE -5 Find the optimal strategies by using the dominance rule for Player A and B in the following game. Also obtain the value of the game.

A's Strategy	B's strategy		
	B1	B2	B3
A1	4	-1	5
A2	0	5	3
A3	3	3	7

Solution:

A's strategy	B's strategy			Row Minima
	B1	B2	B3	
A1	4	-1	5	-1
A2	0	5	3	0
A3	3	3	7	3
Column Maxima	4	5	7	

Maximin

Minimax

Since maximin value is not equal to minimax value the saddle point does not exist. Hence, we are supposed to use dominance rule to reduce the size of the given matrix.

From Player B's perspective: we find Strategy B3 is less attractive or inferior to strategy B1 as each element of B1 is greater than (or equal to) corresponding element of B1. Hence, B3 is eliminated. Then the reduced matrix will be as follows:

A's strategy	B's strategy	
	B1	B2
A1	4	-1
A2	0	5
A3	3	3

From Player A's perspective, none of the strategies are strictly dominated to each other. In this case we identify the strategy which is inferior to other two strategies.

Here, we see the relation, $A_3 \geq \frac{A_1 + A_2}{2}$, so Strategy A3 dominates the convex linear combination of Pure strategies A1 and A2, then we may delete anyone in the combination.

Here, we delete each one of them and generate solution.

By deleting A1, we get

A's Strategy	B's strategy	
	B1	B2
A2	0	5
A3	3	3

Let's check whether saddle point exists or not:

A's Strategy	B's strategy		Row minima
	B1	B2	
A2	0	5	0
A3	3	3	3
Column Maxima	3	5	

Maximax

Minimax

Since maximin value = minimax value, the saddle point exists.

Hence, optimal strategy is (A3, B1), And Value of the game (V) = 3.

By deleting A2, we get

A's Strategy	B's strategy	
	B1	B2
A1	4	-1
A3	3	3

Let's check whether saddle point exists or not:

A's Strategy	B's strategy		Row minima
	B1	B2	
A1	4	-1	-1
A3	3	3	3
Column Maxima	3	5	

Maximin

Minimax

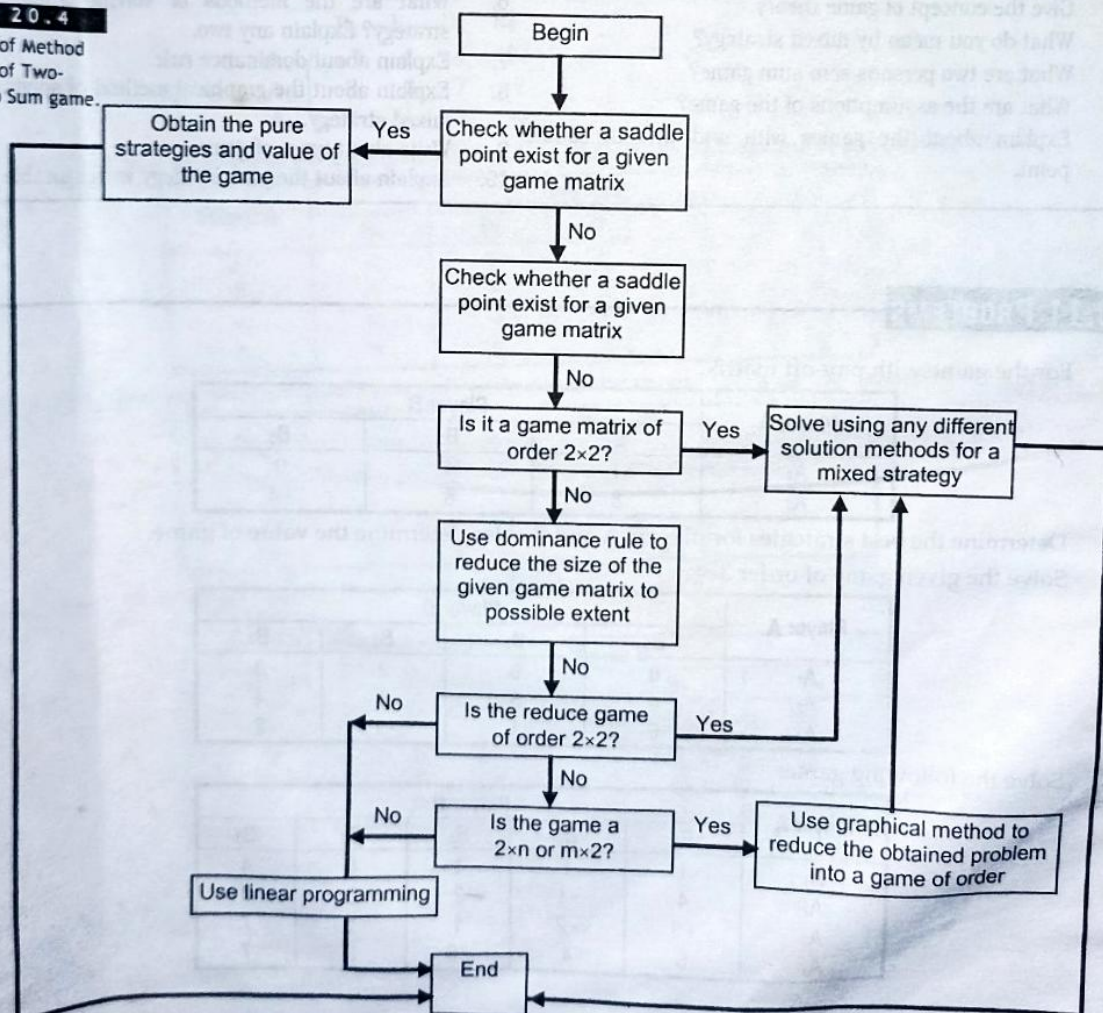
Since maximin value = minimax value, the saddle point exists.

Hence, optimal strategy is (A3, B1)

And Value of the game (V) = 3.

FIGURE 20.4

Flow Chart of Method of Solution of Two-Person Zero Sum game.



Source: Adopted from Sunity Shrestha and others "Critical Thinking & Decision Making" pg. 145-148

20.9 CONCEPT REVIEW

- **Game theory.** It is a peculiar type of decision theory in which one's choice of strategy is determined after considering the possible alternatives available to an opponent playing the same game, rather than just by the possibilities of several outcomes.
- **Players.** The autonomous decision makers in the game are called players.
- **Two-person game.** A game is said to be two-person game if only two players are involved in it.
- **Optimal strategy.** Optimal strategy means the complete plan adopted by a player, which optimizes his gains or losses without knowing the competitors plans.
- **Pure strategy.** Pure strategy refers to the campaign plan for the entire game taking into consideration of possible response to every eventuality.
- **Mixed strategy.** Mixed strategies are those employed strategies in the game in which the interest of the players are not conceivable.
- **Saddle point.** The point in which maximin value of winner is equal to minimax value of loser. The point is also known as equilibrium point and their said to have existence of strategy point.
- **Dominance rule.** This rule is used to reduce the size of the game matrix. It is based on the principle that we determine the strategy, which is preferable to some other strategy/strategies.

20.10 EXERCISE

1. Give the concept of game theory.
2. What do you mean by mixed strategy?
3. What are two persons zero sum game?
4. What are the assumptions of the game?
5. Explain about the games with and without saddle point.
6. What are the methods of solving game of mixed strategy? Explain any two.
7. Explain about dominance rule.
8. Explain about the graphical method of solving game of mixed strategy.
9. Write about types of game.
10. Explain about the pure strategy with suitable example.

20.11 PROBLEMS

1. For the game with pay-off matrix:

Player A	Player B		
	B ₁	B ₂	B ₃
A ₁	1	4	0
A ₂	8	6	-4

Determine the best strategies for players A and B. Also determine the value of game.

2. Solve the given game of order 3×4 .

Player A	Player B			
	B ₁	B ₂	B ₃	B ₄
A ₁	0	6	2	3
A ₂	4	5	3	4
A ₃	6	1	-1	2

3. Solve the following game:

Player A	Player B				
	B ₁	B ₂	B ₃	B ₄	B ₅
A ₁	1	1	1	6	4
A ₂	4	-1	-2	3	3
A ₃	-3	-2	1	1	7
A ₄	6	4	-3	3	7

4. Obtain the pure strategies of the players and also find the value of game.

Player A	Player B			
	B ₁	B ₂	B ₃	B ₄
A ₁	3	-5	0	6
A ₂	-4	-2	1	2
A ₃	5	4	2	3

5. Solve the following game:

Player A	Player B		
	B ₁	B ₂	B ₃
A ₁	-3	14	-3
A ₂	-6	-7	-5
A ₃	-6	19	-9

6. Solve the game:

Player A	Player B			
	B ₁	B ₂	B ₃	B ₄
A ₁	-3	5	3	12
A ₂	7	7	6	8
A ₃	6	0	2	-3

7. Solve the game:

Player A	Player B			
	B ₁	B ₂	B ₃	B ₄
A ₁	-2	6	4	13
A ₂	8	8	7	9
A ₃	7	1	3	-2

8. Identify the pair strategies of the player. Also obtain the value of game and interpret the result.

Player A	Player B				
	B ₁	B ₂	B ₃	B ₄	B ₅
A ₁	4	0	5	7	8
A ₂	0	9	3	5	13
A ₃	17	9	7	15	13
A ₄	2	12	-3	3	2

9. Find the range of values of m and n which will render the entry (2, 2) a saddle point for the game:

Player A	Player B		
	B ₁	B ₂	B ₃
A ₁	0	2	3
A ₂	8	5	n
A ₃	2	m	4

10. Solve the game whose pay-off matrix is given below:

(a)

Player A	Player B			
	B ₁	B ₂	B ₃	B ₄
A ₁	5	4	6	2
A ₂	5	6	4	6
A ₃	6	4	6	2
A ₄	2	6	2	10

(b) Player A $\begin{bmatrix} 2 & -2 & 3 \\ -3 & 5 & -1 \end{bmatrix}$

(c) Player A

	Player B	
	B ₁	B ₂
A ₁	-6	7
A ₂	4	-5
A ₃	-1	-2
A ₄	-2	5
A ₅	7	-6

11. Two breakfast food manufacturers, ABC and XYZ are competing for an increased market share. The pay-off matrix, shown in the following table, describes the increase in market share for ABC and decrease in market share of XYZ.

ABC	XYZ			
	Give Coupons (B ₁)	Decrease Price (B ₂)	Maintain Present Strategy (B ₃)	Increase Advertising (B ₄)
Give coupons (A ₁)	7	3	9	6
Decrease price (A ₂)	11	6	17	8
Maintain present strategy (A ₃)	2	7	5	11
Increase advertising (A ₄)	7	2	12	6

Determine optimal strategies for both the manufacturers and the value of the game.

12. Two competitors are competing for the market share of the similar product. The pay-off matrix in terms of their advertising plan is shown below:

Competitor A	Competitor B		
	No advertising (B ₁)	Medium Advertising (B ₂)	Heavy Advertising (B ₃)
No advertising (A ₁)	8	3	-4
Medium advertising (A ₂)	11	10	13
Heavy advertising (A ₃)	14	12	8

Suggest optimal strategies for the two firms and the net outcome thereof.

13. Solve the following game after reducing it to a 2×2 game

Player A	Player B		
	B ₁	B ₂	B ₃
A ₁	6	12	7
A ₂	11	7	12
A ₃	10	6	11

14. Following is the game problem matrix:

		Company B			
		B ₁	B ₂	B ₃	B ₄
Company A	A ₁	40	30	24	70
	A ₂	50	28	16	20
	A ₃	80	4	38	10
	A ₄	-10	8	22	0

Determine the optimum strategies for the Company A and Company B. What is the value of the game?

15. Solve the game whose pay-off matrix is given below: -2

		Firm B			
		B ₁	B ₂	B ₃	B ₄
Firm A	A ₁	3	2	4	0
	A ₂	3	4	2	4
	A ₃	4	2	4	0
	A ₄	0	4	0	8

16. A steel company is negotiating with its union for revision of wages to its employees. The management, with the help of a mediator, has prepared matrix shown below. Plus sign represents wage increase, while negative sign stands for wage decrease. Union has also constructed a table which is comparable to that developed by the management. The management does not have the specific knowledge of game theory of game to select the best strategy or strategies for the firm. You have to assist the management on the problem. What game value and strategies are available to the opposing group?

Additional Cost to steel Company (Rs.)

		Union Strategy			
		U ₁	U ₂	U ₃	U ₄
Company Strategy	C ₁	2.50	2.70	3.50	0.20
	C ₂	2.00	1.60	0.80	0.80
	C ₃	1.40	1.20	1.50	1.30
	C ₄	3.00	1.40	1.90	0.00

17. Two players A and B match coins. If the coin matches, then A wins one unit of value, if the coins do not match, then B wins one unit of value. Determine optimal strategies for the players and the value of game.

Answers

1. A₁, B₃ value of the game = 0
2. A₁, B₃, V = 3
3. A₁, B₃, V = 1
4. A₃, B₃, V = 1
5. A₁, B₁: A₁, B₃, V = -3
6. A₂, B₃, V = 6
7. A₂, B₃ and value of the game = 7
8. A₃, B₃, V = 7; Interpretation: By employing A₃ strategy by A will return him seven in the long-run of the game while by employing B₃ by player B will lose the value seven in the long-run of the game
9. $m \leq 5, n > 5$
10. (a) $S_A = \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$; $S_B = \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$ V = $\frac{14}{3}$
 (b) $S_A = \left(\frac{2}{3}, \frac{1}{3}\right)$ $S_B = \left(\frac{7}{12}, \frac{5}{12}, 0\right)$ V = $\frac{1}{3}$ (c) $S_A = \left(0, 0, 0, \frac{13}{20}, \frac{7}{20}\right)$ $S_B = \left(\frac{11}{20}, \frac{9}{20}\right)$ V = $\frac{23}{20}$
11. Optimal strategies of ABC = $\left(0, \frac{1}{2}, \frac{1}{2}, 0\right)$ and optimal strategies of XYZ = $\left(\frac{1}{10}, \frac{9}{10}, 0, 0\right)$ and Value of the game = $\frac{13}{2}$
12. Optimal strategies of A = $\left(0, \frac{4}{7}, \frac{3}{7}\right)$ and optimal strategies of B = $\left(0, \frac{5}{7}, \frac{2}{7}\right)$ and Value of the game = $\frac{76}{7}$
13. Optimal strategies of A = $\left(\frac{2}{5}, \frac{3}{5}, 0\right)$ and optimal strategies of B = $\left(\frac{1}{2}, \frac{1}{2}, 0\right)$ and Value of the game = 9
14. Optimal strategies for Company A = $\left(\frac{17}{20}, 0, \frac{3}{20}, 0\right)$, Optimal strategies for Company B = $\left(0, \frac{7}{20}, \frac{13}{20}, 0\right)$ and Value of the game 26.1
15. Optimal strategies for Firm A = $\left(\frac{2}{3}, \frac{1}{3}, 0, 0\right)$, Optimal strategies for Firm B = $\left(0, 0, \frac{2}{3}, \frac{1}{3}\right)$ and Value of the game $\frac{11}{3}$
16. $P(U_1) = 7/13, P(U_2) = 6/13; P(C_1) = 1/13, P(C_2) = 12/13$
17. $S_A = \left(\frac{1}{2}, \frac{1}{2}\right)$ $S_B = \left(\frac{1}{2}, \frac{1}{2}\right)$ V = 0